



- (7)





4. A small brick of mass 0.5 kg is placed on a rough plane which is inclined to the horizontal at an angle  $\theta$ , where  $\tan \theta = \frac{4}{3}$ , and released from rest. The coefficient of friction between the brick and the plane is  $\frac{1}{3}$ .

Find the acceleration of the brick.

(9)

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.







8. [In this question  $\mathbf{i}$  and  $\mathbf{j}$  are horizontal unit vectors due east and due north respectively.]

A hiker  $H$  is walking with constant velocity  $(1.2\mathbf{i} - 0.9\mathbf{j}) \text{ m s}^{-1}$ .

- (a) Find the speed of  $H$ .

(2)

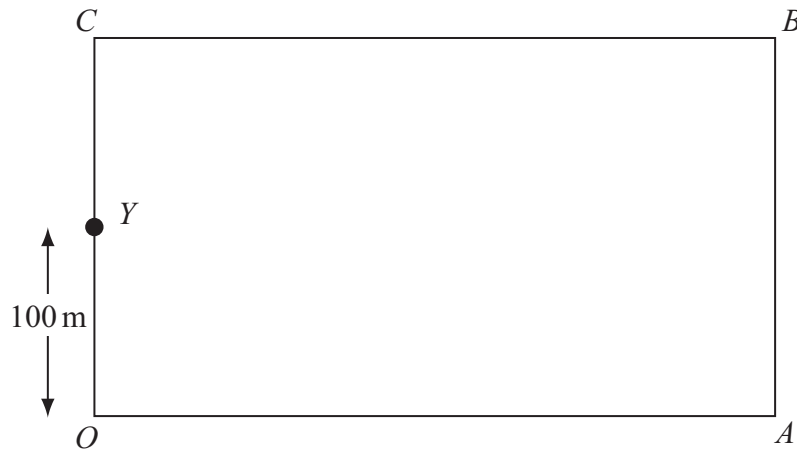


Figure 3

A horizontal field  $OABC$  is rectangular with  $OA$  due east and  $OC$  due north, as shown in Figure 3. At twelve noon hiker  $H$  is at the point  $Y$  with position vector  $100\mathbf{j}$  m, relative to the fixed origin  $O$ .

- (b) Write down the position vector of  $H$  at time  $t$  seconds after noon.

(2)

At noon, another hiker  $K$  is at the point with position vector  $(9\mathbf{i} + 46\mathbf{j})$  m. Hiker  $K$  is moving with constant velocity  $(0.75\mathbf{i} + 1.8\mathbf{j}) \text{ m s}^{-1}$ .

- (c) Show that, at time  $t$  seconds after noon,

$$\overrightarrow{HK} = [(9 - 0.45t)\mathbf{i} + (2.7t - 54)\mathbf{j}] \text{ metres.}$$

(4)

Hence,

- (d) show that the two hikers meet and find the position vector of the point where they meet.

(5)

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[illegible]